CUTTING A PSEUDO-COHERENT COMPLEX IN TWO PIECES AFTER LOCALIZING, ONCE A COHOMOLOGY GROUP IS GENERATED BY ITS REDUCTION AT A PRIME

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Editorial note. The proof refers once to a statement of this chapter and twice to results of other chapters, namely the localization statement for cokernels and the distinguished triangle of truncations; all three are carried as navigation stubs with their roles recorded in the reasons.

Editorial note. The statement and proof use the truncation notation tau of the shared preamble, which the retained preamble supplies verbatim.

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Allowed standard prerequisites (editorial).

- A pseudo-coherent complex is represented by a bounded above complex of finite free modules, and a finite module over a local ring that is generated by its reduction can be generated by lifts modulo an arbitrarily high power of the maximal ideal. used at proof lines 19767-19768 and 19786-19794. Justifies the choice of the elements y_b and the localization that makes the three listed conditions hold.
- If a map of finite free modules has a surjective reduction modulo a prime and its cokernel becomes finite projective after inverting some element outside the prime, then the cokernel itself is finite projective after that localization. invoked at proof lines 19793-19799. Produces the finite projective quotient E^{i+1} that makes the upper truncation perfect.
- For a distinguished triangle whose first term has no cohomology above a given degree and whose third term is perfect with amplitude starting at that degree, the triangle splits and exhibits the middle term as a direct

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sum of the two truncations. invoked at proof lines 19810-19818. Gives the displayed canonical isomorphism and finishes the proof.

Source reference. Original lemma 00O0: sources/stacks-chow/tags.txt, source lines 878–878. The reference is the localization statement applied at proof lines 19793-19794 to obtain the three conditions after inverting an element outside the prime; the three conditions and their use are inside the counted span.

Source reference. Original lemma 0BCG: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19704–19704. The reference is applied at proof lines 19811-19812 to deduce the canonical isomorphism from the distinguished triangle; the identification of the upper truncation as perfect is inside the counted span.

Source reference. Original remark 08J5: sources/stacks-chow/tags.txt, source lines 10838–10838. The reference is the standard distinguished triangle relating a complex to its two truncations, cited at proof lines 19817-19818; it is a navigation pointer to the triangle and the splitting statement that consumes it is inside the counted span.

Lemma 0.2. *Let R be a ring. Let $\mathfrak{p} \subset R$ be a prime ideal. Let $K^\bullet$ be a pseudo-coherent complex of R-modules. Assume that for some $i \in \mathbf{Z}$ the map*

$$H^i(K^\bullet) \otimes_R \kappa(\mathfrak{p}) \longrightarrow H^i(K^\bullet \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p}))$$

is surjective. Then there exists an $f \in R$, $f \notin \mathfrak{p}$ such that $\tau_{\geq i+1}(K^\bullet \otimes_R R_f)$ is a perfect object of $D(R_f)$ with tor amplitude in $[i+1, \infty]$ and a canonical isomorphism

$$K^\bullet \otimes_R R_f \cong \tau_{\leq i}(K^\bullet \otimes_R R_f) \oplus \tau_{\geq i+1}(K^\bullet \otimes_R R_f)$$

in $D(R_f)$.

Proof. In this proof all tensor products are over R and we write $\kappa = \kappa(\mathfrak{p})$. We may assume that $K^\bullet$ is a bounded above complex of finite free R-modules. Let us inspect what is happening in degree i:

$$\ldots \to K^{i-1} \xrightarrow{d^{i-1}} K^i \xrightarrow{d^i} K^{i+1} \to \ldots$$

Let $0 \subset V \subset W \subset K^i \otimes \kappa$ be defined by the formulas

$$V = \operatorname{Im}\left(K^{i-1} \otimes \kappa \to K^i \otimes \kappa\right) \quad \text{and} \quad W = \operatorname{Ker}\left(K^i \otimes \kappa \to K^{i+1} \otimes \kappa\right)$$

Set $\dim(V) = r$, $\dim(W/V) = s$, and $\dim(K^i \otimes \kappa/W) = t$. We can pick $x_1, \ldots, x_r \in K^{i-1}$ which map by d^{i-1} to a basis of V. By our assumption we can pick $y_1, \ldots, y_s \in \operatorname{Ker}(d^i)$ mapping to a basis of W/V. Finally, choose $z_1, \ldots, z_t \in K^i$ mapping to a basis of $K^i \otimes \kappa/W$. Then we see that the elements $d^i(z_1), \ldots, d^i(z_t) \in K^{i+1}$ are linearly independent in $K^{i+1} \otimes_R \kappa$. By Algebra, Lemma 00O0 we may after replacing R by R_f for some $f \in R$, $f \notin \mathfrak{p}$ assume that

(1) $d^i(x_a), y_b, z_c$ is an R-basis of K^i,
(2) $d^i(z_1), \ldots, d^i(z_t)$ are R-linearly independent in K^{i+1}, and
(3) the quotient $E^{i+1} = K^{i+1}/\sum Rd^i(z_c)$ is finite projective.

Since d^i annihilates $d^{i-1}(x_a)$ and y_b, we deduce from condition (2) that $E^{i+1} = \operatorname{Coker}(d^i : K^i \to K^{i+1})$. Thus we see that

$$\tau_{\geq i+1}K^\bullet = (\ldots \to 0 \to E^{i+1} \to K^{i+2} \to \ldots)$$

is a bounded complex of finite projective modules sitting in degrees $[i+1, b]$ for some b. Thus $\tau_{\geq i+1}K^\bullet$ is perfect of amplitude $[i+1, b]$. Since $\tau_{\leq i}K^\bullet$ has no cohomology in degrees $> i$, we may apply Lemma 0BCG to the distinguished triangle

$$\tau_{\leq i}K^\bullet \to K^\bullet \to \tau_{\geq i+1}K^\bullet \to (\tau_{\leq i}K^\bullet)[1]$$

(Derived Categories, Remark 08J5) to conclude. □